Phase IX final technical closure gate

Amos Jay Maley

Independent Researcher, Pho Thong, Ang Thong, Thailand[a)]

(Dated: 4 April 2026)

This packet executes Phase IX of the clay-closure upgrade plan on top of the hardened proof surface produced by Phases I–VIII. It assembles, in one synchronized audit document, the final pass conditions, the load-bearing theorem-node closure matrix, the submission-package inventory, the final sign-off checklist, and the blocker-by-blocker closure memo. The packet adds no mathematics and is not a proof home. Its purpose is narrower: it makes explicit the internal package-level decision rule that must be satisfied before the manuscript suite is handed to outside referees and specialist auditors.

[a)]Electronic mail: somamaley@gmail.com

I. SCOPE, USE RULE, AND CLOSURE STATE AT ENTRY TO PHASE IX

This packet is the project-internal final technical-closure gate required by Phase IX of the upgrade plan. It does not alter the mathematics already carried by the hardened live proof surface. Instead it assembles, in one place, the final pass conditions, the theorem-node closure matrix, the submission-package inventory, the sign-off checklist, and the closure memo that records why each previously identified blocker is now closed.

Live proof burden. The live proof burden remains exactly the four proof-home roots `core.tex`, `companion1.tex`, `companion2.tex`, and `companion3.tex`, together with the theorem-facing appendices compiled inside those roots. The proof kernel `proofkernel.tex` and this Phase IX packet are synchronized audit documents only: they add no mathematics and do not change the proof burden.

Use rule. A referee-grade read should proceed in the following order. First attack the core paper or the proof kernel. Second, if a live theorem node needs more detail, descend only to the explicit proof home cited for that theorem. Third, consult this Phase IX packet only for the final closure matrix, submission inventory, and checklist. Reserve routing notes remain non-mathematical and should never be required to verify a live theorem edge.

Closure state at entry. By the time this packet is assembled, the earlier phases have already frozen the theorem surface, repaired the known seam defects, installed theorem-local ledgers, hardened quantifiers and domain transport, exposed the functional / measure backbone, extracted the proof kernel, narrowed the claim framing to the exact local-net / vacuum-sector endpoint, and completed the manuscript-side adversarial-verification pass. The only remaining internal task is therefore to record explicitly that the packaged suite now satisfies the required Phase IX pass conditions in one synchronized place.

II. REQUIRED PASS CONDITIONS AT THE FINAL GATE

Theorem-local closure (✓):

Local dependency ledgers now sit at the critical core, Companion I, Companion II, and Companion III nodes, and the proof-home tables plus packet docket identify exactly where each live theorem is proved and what consumes it next.

No seam ambiguity (✓):

The role split among 5.74, 5.75, 5.76, 5.77, F.213, F.214, and F.216 is synchronized in the core, the proof homes, the phase-freeze concordance appendices, and the proof kernel. The finite-cap bridge into 5.74 is theorem-facing.

Quantifier discipline (✓):

Capwise versus full-union claims, dyadic versus all-time transport, finite-cap graph-core approximation, and logarithmic-depth Wilson-return uniformity are all frozen in theorem-facing form on the live route.

Imported background bounded (✓):

The standard KP/Dobrushin and OS/Wightman inputs are boxed explicitly in the concordance and proof-kernel imports appendices, and no other mathematical outsourcing is claimed.

Claim hygiene (✓):

The title, abstract, main theorem section, endpoint section, proof kernel, and Companion III all frame exactly the bounded-region local gauge-invariant sharp-local net and its vacuum-sector mass gap, while preserving the qualitative faithful-Wilson universality claim and the explicit quantitative non-claims.

Independent auditability (✓):

The proof kernel now exposes the packet route, seam details, import boundary, and exclusions boundary in a form that outside specialists can attack directly without reconstructing the proof from reserve notes.

Interpretation. At this gate the project is not claiming that outside referee validation has already happened. It is claiming the narrower and necessary internal condition: after the synchronized repairs of Phases I–VIII, the packaged suite now exposes the theorem object, the proof homes, the import boundary, the seam roles, and the audit documents in a form that an outside expert can attack directly.

III. CLOSURE MATRIX BY LOAD-BEARING THEOREM NODE

Core Theorem I.1 — passed:

Exact endpoint wording, qualitative faithful-Wilson universality, and the explicit non-claims are frozen against overreading. *Witness:* Core Sections I and VII; proof-kernel Sections 1 and 7.

Core Section II packet map — passed:

Packet 8 now visibly consumes the Lane A seam nodes and the public spine no longer hides that dependence. *Witness:* Core Section II and Appendix A; proof-kernel dependency-spine section.

IV.2 / III.17 — passed:

Same-scale Wilson return is synchronized everywhere at finite logarithmic depth rather than vague fixed depth. *Witness:* Core Section IV; Companion I theorem-local note at III.17.

V.6 / 5.69 — passed:

Finite-truncation inverse control carries an explicit downstream-use ledger into the finite-cap bridge and closure packet. *Witness:* Core Section V; Companion II theorem-local note at 5.69.

V.9 / F.331 — passed:

The mixed-correlator closure node no longer floats in isolation; it now feeds a theorem-facing bridge into 5.74. *Witness:* Core Section V; Companion II state-construction ledger and bridge notes.

V.10 / 5.74 — passed:

Finite-cap state existence, positivity, unitality, and OS structure are theorem-faced locally instead of reconstructed from distributed hints. *Witness:* Core Section V; Companion II theorem 5.74 together with the theorem-facing bridge notes.

V.11 / 5.76 — passed:

The inductive-union passage remains sharply separated from 5.75 compatibility language across all proof homes and summaries. *Witness:* Core

Section V; Companion II theorem-local note at 5.76; freeze appendix crosswalk.

V.12 / 5.77 — passed:

Bounded-base cyclicity remains a separate theorem-facing output that is consumed transparently at QE3 and reconstruction. *Witness:* Core Section V; Companion II theorem-local note at 5.77.

F.212 / VI.2 — passed:

The QE3 seam theorem now cites the correct nodes with the explicit local split 5.75 / 5.74(b) / 5.76 / F.213 / F.214. *Witness:* Core Section VI; Companion I theorem F.212 and its local note.

F.216 / VI.5 — passed:

Continuum vacuum-gap transport keeps the density / graph-core handoff explicit and does not hide a domain argument in prose. *Witness:* Core Section VI; Companion I theorem-local note at F.216.

M.1 / VII.1 — passed:

The Euclidean OS dossier remains the exact internal endpoint handoff, with the imported standard background isolated rather than smeared. *Witness:* Core Section VII; Companion III theorem M.1; proof-kernel imported-background appendix.

O.3 / VII endpoint — passed:

Qualitative faithful-Wilson universality remains visibly distinct from the ρ-indexed quantitative non-claim and from any nonlocal-sector overreading. *Witness:* Core Section VII; Companion III endpoint section; proof-kernel Section VII.

The purpose of this matrix is not to restate every proof. It is to record that each load-bearing node singled out in the closure plan now has a theorem-facing witness inside the packaged suite and therefore no longer needs reconstructive labor to see why it sits on the public spine.

IV. SUBMISSION PACKAGE ASSEMBLED AT PHASE IX

`core.tex` — live:

Clean submission root stating the theorem object, the ten-packet public spine, the exact endpoint, and the theorem-facing appendices.

`companion1.tex` — live:

Proof home for the ultraviolet / Route 1 / QE3 transport chain, including the repaired seam theorem F.212 and the continuum-gap packet.

`companion2.tex` — live:

Proof home for the Lane A finite-cap closure chain, the state-construction ledger, and the theorem-facing positivity bridge into 5.74.

`companion3.tex` — live:

Proof home for reconstruction, non-triviality, qualitative faithful-Wilson universality, and the exact local-net endpoint.

`proofkernel.tex` — derived audit:

Optional expert-circulation kernel containing the load-bearing packet route, the imported-background appendix, the exclusions appendix, and the synchronized audit pointers.

Phase VIII verification dossier — derived audit:

The synchronized manuscript-side adversarial-verification reports, issue tracker, dispositions, and sign-off checklist retained at the repository root and in the core / kernel verification appendices.

`phase9_closure_gate.tex` — derived audit:

Final technical-closure packet collecting the pass conditions, theorem-node closure matrix, package inventory, checklist, and closure memo.

Reserve routing notes — reserve:

Retained routing or editorial material only, including the synchronized support roots for the analytic certification dossier, the referee concordance and reading note, and the supplementary referee dossier, together with the

reserve referee-audit packet, the executive proof sketch, and supporting editorial notes. These files add no mathematics and are not required to verify any live theorem edge.

Clean-build rule. The packaged project remains configured for pdfLaTeX / `latexmk` with no BibTeX step. The shared manual bibliography block in `shared/bibliography.tex` remains authoritative for all roots. The external-label include files stay synchronized so the core, companions, and kernel can be built independently on Overleaf.

V. FINAL SIGN-OFF CHECKLIST AND DECISION RULE

1. Every theorem on the public spine is attached to a single canonical proof home. *Primary evidence:* Core Appendix C and the companion proof-home guides. ✓
2. The role split among 5.74, 5.75, 5.76, 5.77, F.213, F.214, and F.216 is identical in the core, Companion I, Companion II, the proof kernel, and the reserve-routing notices. *Primary evidence:* Core Sections V–VI; companion notes; kernel Sections V–VI. ✓
3. F.212 cites the correct nodes with no ambiguity. *Primary evidence:* Companion I theorem F.212; core Section VI. ✓
4. The finite-cap positivity / unitality bridge is theorem-facing and cited directly by 5.74. *Primary evidence:* Companion II theorem-facing bridge notes and theorem 5.74; core Section V. ✓
5. All fixed-cap, familywise, dyadic, and full-union distinctions are explicit. *Primary evidence:* Core Sections IV–VI; Companion II capwise ledgers; kernel packet route. ✓
6. Imported standard results are isolated and declared only where used. *Primary evidence:* Core Appendix B; proof-kernel imported-background appendix. ✓
7. The title, abstract, theorem statement, and endpoint section match the actual theorem object exactly. *Primary evidence:* Core front matter and Sections I, VII; proof-kernel front matter. ✓
8. A specialist can audit the proof kernel without reserve materials. *Primary evidence:* Proof-kernel root, reading guide, exclusions appendix, and the synchronized Phase VIII

/ Phase IX audit pointers. ✓

9. Every previous red-team issue has a manuscript-level disposition. *Primary evidence:* The retained Phase VIII tracker, disposition memo, sign-off checklist, and the synchronized verification appendices in the core and kernel. ✓

Decision rule. All nine mandatory checklist items are now marked yes on the packaged suite. Therefore, conditional on the underlying mathematics being correct, the manuscript bundle has reached the internal technical-closure state targeted by the plan: the remaining barriers are outside validation, top-tier journal review, post-publication scrutiny, and the waiting-period realities attached to any millennium-level claim.

VI. FINAL CLOSURE MEMO

Blocker-by-blocker closure record.

B1. The QE3 seam theorem F.212 no longer conflates 5.75 with 5.76. Its proof and theorem-local note now separate bounded-state restriction compatibility, finite-cap sharp-local OS structure, the inductive-union passage, and the density / graph-core handoff. The original closure-critical citation defect has therefore been removed at the proof home itself.

B2. The finite-cap bridge into 5.74 is now theorem-facing. The state-construction ledger and the local bridge note make positivity, unitality, bounded positive-time representatives, and the finite-cap OS mechanism explicit where 5.74 uses them. The earlier objection that the theorem-facing bridge looked weaker than the theorem it served is therefore closed.

B3. Packet 8 is no longer compressed in a way that forces the reader to infer hidden Lane A dependence. The core packet map, the QE3 section, and the proof kernel all state directly that the dyadic-to-all-time transport packet consumes the repaired Lane A seam nodes. The distributed-hint failure mode is therefore closed.

B4. The theorem numbering and routing surface are now frozen. The packet docket, proof-home tables, canonical numbering crosswalk, and synchronized phase-freeze appendix make it possible to answer, for every public-spine theorem, where it is proved, what it does, and what consumes it next. The numbering-friction blocker is therefore closed.

B5. The front matter and endpoint prose no longer invite a broader reading than the theorem supports. The title, abstract, main theorem section, proof kernel, and Companion III

now all state the bounded-region local gauge-invariant sharp-local net / vacuum-sector mass-gap endpoint with the qualitative faithful-Wilson claim and the explicit quantitative / nonlocal non-claims preserved. The framing blocker is therefore closed.

B6. The project now contains a standalone proof kernel exposing the live packet route, seam details, import boundary, exclusions boundary, and expert reading guide without requiring reserve dossiers. This does not replace later external audit, but it does close the internal blocker that previously forced specialists to reconstruct the proof route from multiple surfaces.

Remaining work after Phase IX. Within the manuscript bundle, no further theorem-local closure repair is planned. The remaining steps are external: independent specialist reading, top-tier referee review, post-publication scrutiny, and the real-world waiting-period constraints attached to any millennium-level claim. Within that scope, the suite has passed the Phase IX final technical-closure gate.

REFERENCES

[1]K. Osterwalder and R. Schrader, *Axioms for Euclidean Greenś functions*, *Communications in Mathematical Physics* **31** (1973).

[2]K. Osterwalder and R. Schrader, *Axioms for Euclidean Greenś functions II*, *Communications in Mathematical Physics* **42** (1975).

[3]K. Osterwalder and E. Seiler, *Gauge field theories on a lattice*, *Annals of Physics* **110** (1978).

[4]K. G. Wilson, *Confinement of quarks*, *Physical Review D* **10** (1974).

[5]D. J. Gross and F. Wilczek, *Ultraviolet behavior of non-Abelian gauge theories*, *Physical Review Letters* **30** (1973).

[6]H. D. Politzer, *Reliable perturbative results for strong interactions?*, *Physical Review Letters* **30** (1973).

[7]M. Lüscher, *Properties and uses of the Wilson flow in lattice QCD*, *Journal of High Energy Physics* **1008** (2010) 071.

[8]D. G. Aronson, *Bounds for the fundamental solution of a parabolic equation*, *Bulletin of the American Mathematical Society* **73** (1967).

[9]E. B. Davies, *Heat Kernels and Spectral Theory*, Cambridge University Press, 1989.

[10]R. Kotecký and D. Preiss, *Cluster expansion for abstract polymer models*, *Communications in Mathematical Physics* **103** (1986) 491–498.

[11]D. C. Brydges, *A short course on cluster expansions*, in *Critical Phenomena, Random Systems, Gauge Theories* (Les Houches, Session XLIII, 1984), eds. K. Osterwalder and R. Stora, North-Holland, 1986.

[12]R. L. Dobrushin, *The description of a random field by means of conditional probabilities and conditions of its regularity*, *Theory of Probability and its Applications* **13** (1968) 197–224.

[13]K. Hornik and B. Grün, *Amos-type bounds for modified Bessel function ratios*, *J. Math. Anal. Appl.* **408**(1):91–101, 2013.

[14]K. Hornik and B. Grün, *Generalized Amos–type bounds for modified Bessel function ratios*, *Mathematical Inequalities & Applications* **27**(4) (2024), 775–787.

[15]D. E. Amos, *Computation of modified Bessel functions and their ratios*, *Mathematics of Computation* **28**(125) (1974), 239–251.

[16]P.-O. Dehaye, On an identity due to Bump and Diaconis, and Tracy and Widom, *Canadian Mathematical Bulletin* **54**(2) (2011), 255–269.

[17]D. Bump and P. Diaconis, Toeplitz minors, *J. Combin. Theory Ser. A* **97**(2) (2002), 252–271.

[18]A. Edrei, On the generating functions of totally positive sequences, *Transactions of the American Mathematical Society* **74** (1953), 367–383.

[19]S. Karlin, *Total Positivity*, Vol. 1, Stanford University Press, 1968.

[20]A. Pinkus, *Totally Positive Matrices*, Cambridge University Press, 1996; reprinted with corrections, 2010.

[21]Zs. Páles and É. Székelynéné Radácsi, *A new characterization of convexity with respect to Chebyshev systems*, *Journal of Mathematical Inequalities* **12**(3) (2018), 605–617.

[22]Sz. Wasowicz, *Some properties of generalized higher-order convexity*, *Publ. Math. Debrecen* **68** (2006), no. 1–2, 171–182.

[23]M. Lüscher and P. Weisz, *On-shell improved lattice gauge theories*, *Communications in Mathematical Physics* **97** (1985), 59–77.

[24]N. Husung, *Logarithmic corrections to O(a) and O(a^2) effects in lattice QCD with Wilson or Ginsparg–Wilson quarks*, *European Physical Journal C* **83** (2023), 142.

[25]S. Sugiura, *Fourier series of smooth functions on compact Lie groups*, *Osaka Journal of Mathematics* **8** (1971), 33–47.

[26]A. Jaffe and E. Witten, *Quantum Yang–Mills Theory*, in *The Millennium Prize Problems*, Clay Mathematics Institute/American Mathematical Society, 2006.

[27]S. Doplicher, R. Haag, and J. E. Roberts, *Local observables and particle statistics I*, *Communications in Mathematical Physics* **23** (1971), 199–230.

[28]S. Doplicher, R. Haag, and J. E. Roberts, *Local observables and particle statistics II*, *Communications in Mathematical Physics* **35** (1974), 49–85.

[29]S. Doplicher and J. E. Roberts, *Why there is a field algebra with a compact gauge group describing the superselection structure in particle physics*, *Communications in Mathematical Physics* **131** (1990), 51–107.

[30]R. F. Streater and A. S. Wightman, *PCT, Spin and Statistics, and All That*, Princeton University Press, 2000.

[31]R. Haag, *Local Quantum Physics*, 2nd ed., Springer, 1996.

[32]S. Basu, R. Pollack, and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, 2nd ed., Springer, 2006.